

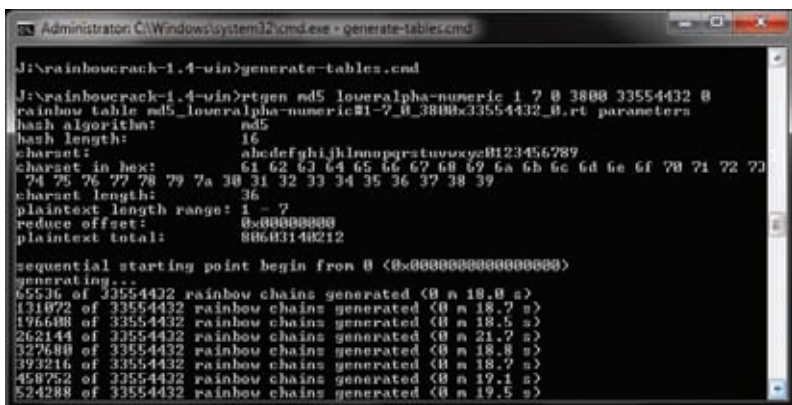
Instead of md5, you could have used any other algorithm here. Rainbowcrack can be extended using dlls to support any hash algorithm.

Here we are generating a rainbow table with the following specifications:

hash_algorithm	md5
charset	loweralpha-numeric = [abcdefghijklmnopqrstuvwxyz0123456789]
plaintext_len_min	1
plaintext_len_max	7
chain_len	3800
chain_num	33554432

This command has to be run 6 times for each table index, from 0 to 5, and each time it will take around 2hrs on a Core i5 system! Each resulting file will be around 512MB, making the whole table collection around 3GB in size.

As you increase the number of characters in the character set and the length, the size will go up drastically.



```

Administrator C:\Windows\system32\cmd.exe - generate-tables.cmd

J:\rainbowcrack-1.4-win>generate-tables.cmd

J:\rainbowcrack-1.4-win>rtgen md5 loweralpha-numeric 1 7 0 3800 33554432 0
rainbow table md5_loweralpha-numeric#1-7_0_3800x33554432_0.rt parameters
hash algorithm:      md5
hash length:         16
charset:              abcdefghijklmnopqrstuvwxyz0123456789
charset in hex:       61 62 63 64 65 66 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73
                    74 75 76 77 78 79 7a 30 31 32 33 34 35 36 37 38 39
charset length:       36
plaintext length range: 1 - 7
reduce offset:        0x00000000
plaintext total:      00603140212

sequential starting point begin from 0 <0x0000000000000000>
generation...
65536 of 33554432 rainbow chains generated <0 m 10.0 s>
131072 of 33554432 rainbow chains generated <0 m 18.7 s>
196608 of 33554432 rainbow chains generated <0 m 18.5 s>
262144 of 33554432 rainbow chains generated <0 m 21.7 s>
327680 of 33554432 rainbow chains generated <0 m 18.8 s>
393216 of 33554432 rainbow chains generated <0 m 18.7 s>
458752 of 33554432 rainbow chains generated <0 m 19.1 s>
524288 of 33554432 rainbow chains generated <0 m 19.5 s>
  
```

Running the rtgen tool to generate rainbow tables

This will result in files named “md5_loweralpha-numeric#1-7_0_3800x33554432_0.rt” to “md5_loweralpha-numeric#1-7_0_3800x33554432_5.rt”

The resulting rainbow tables will not be sorted, and hence will not be easy to search in. So the next step is to sort the tables using the rtsort command. This command simply takes the filename of the unsorted table as a parameter. This takes under a minute to execute.

Since rainbow tables are large, you can also use rt2rtc command to compress the tables into “.rtc” files which can be used with the cracking application without needing to be decompressed.